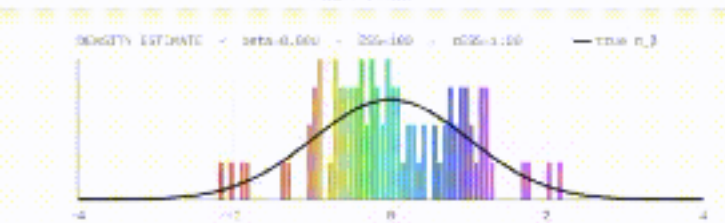
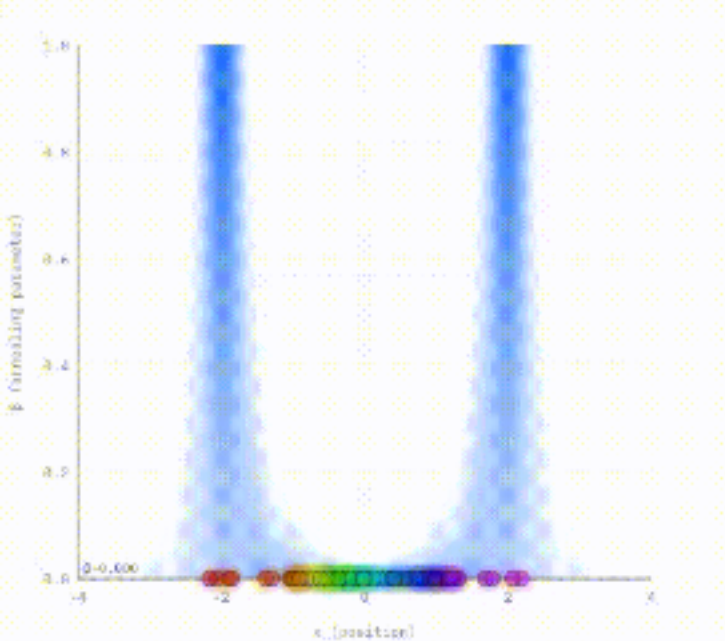


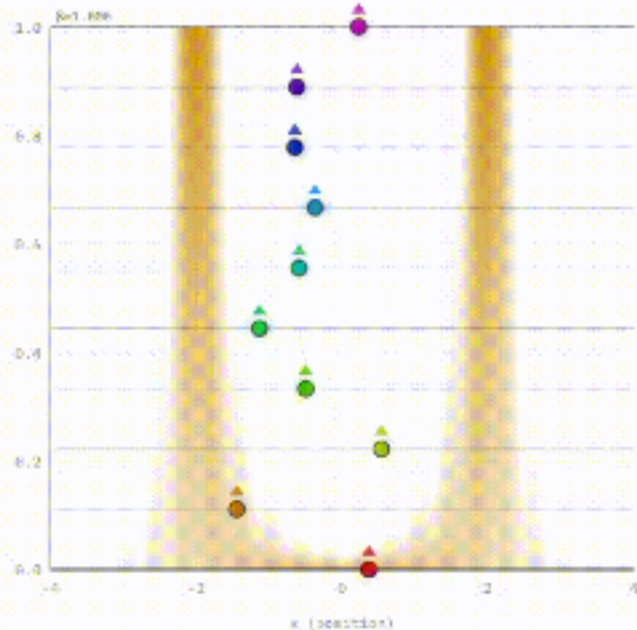
SEQUENTIAL VERSUS PARALLEL TRAINING



Sequential Annearling



β (normalizing parameter)



Index process (last 10 items)

Index



DENSITY ESTIMATE - beta=1 (top level) - RTD

— true n_p



We simulate a system of particles approximating the annealing path

construct path sequentially

▶ samples generated in parallel



construct path in parallel

samples generated sequentially



Interacting $\rightarrow$ Parallel Tempering

► Distributed $\rightarrow$ Simulated sampling

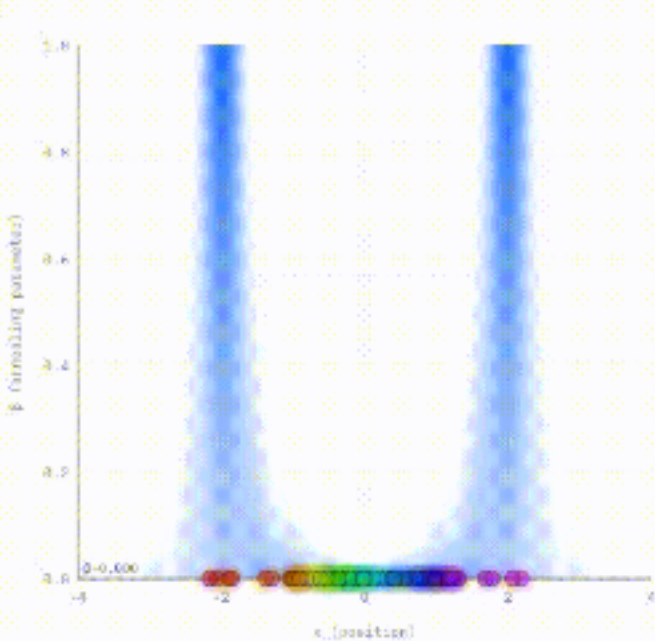
► Interacting $\rightarrow$ Sequential Monte Carlo Samplers

► Distributed $\rightarrow$ Annealed Importance Sampling

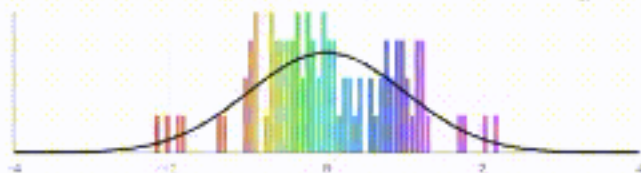


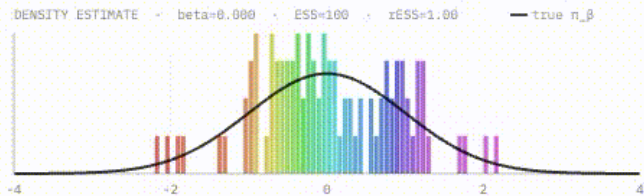
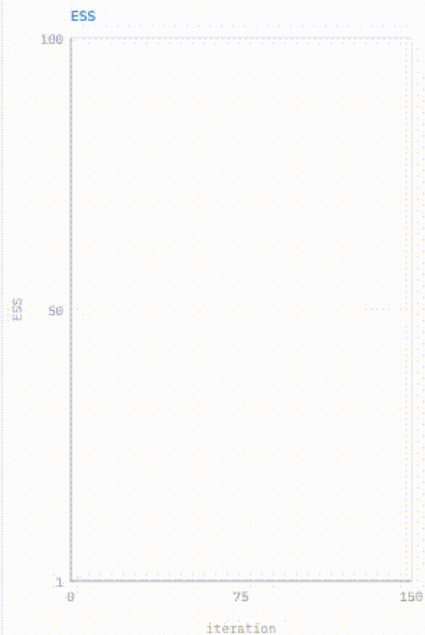
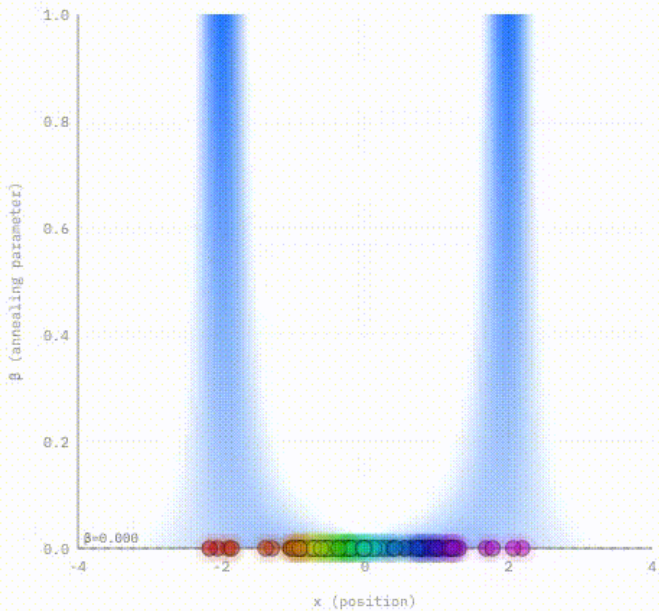


BuilditusingNMCNMC

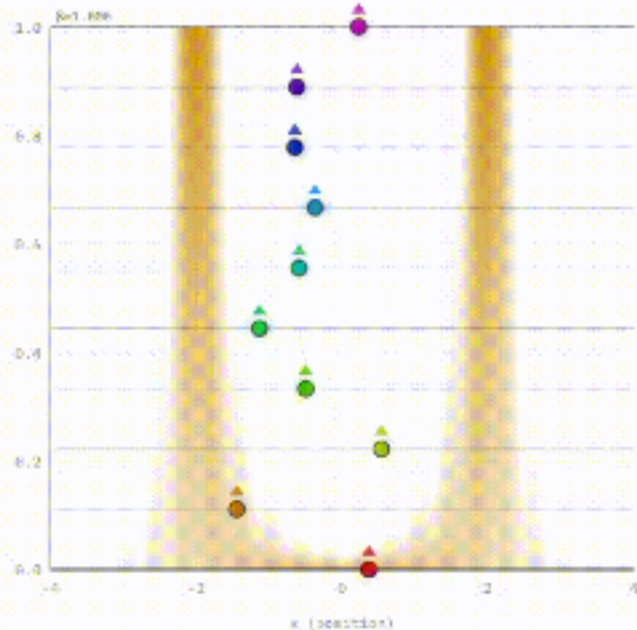


ODSGITV ESTIMATE - beta=0.000 - 200=100 - nSG=1.00 — true n,β





β (normalizing parameter)



Index process (last 10 items)

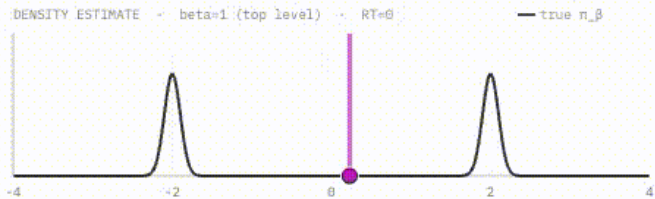
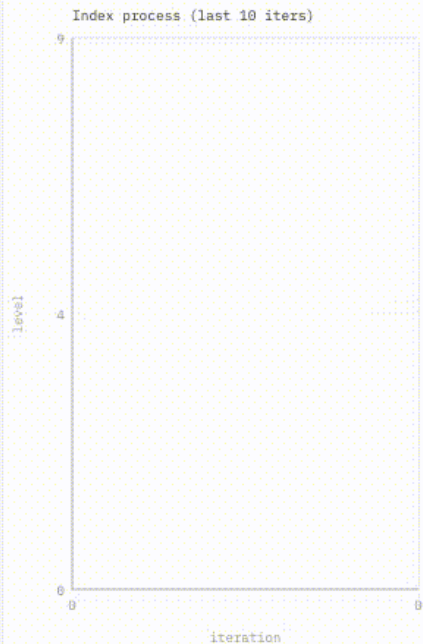
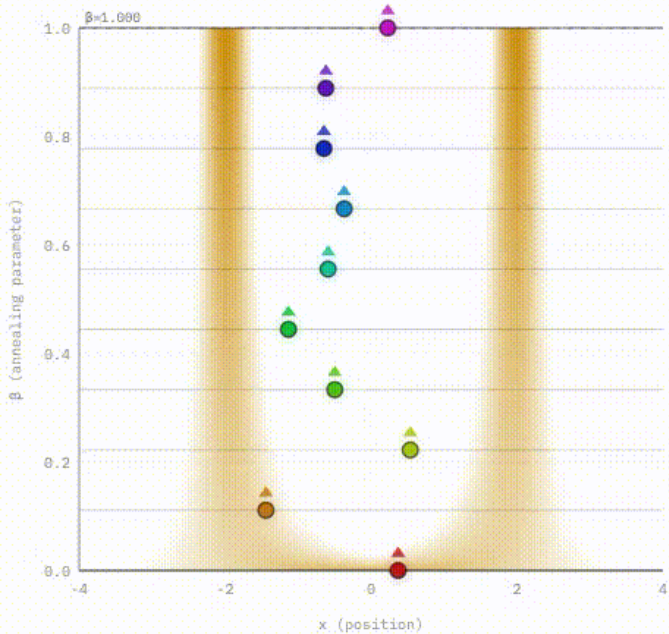
Index



DENSITY ESTIMATE - beta=1.0 (top level) - RTD

— true n_p



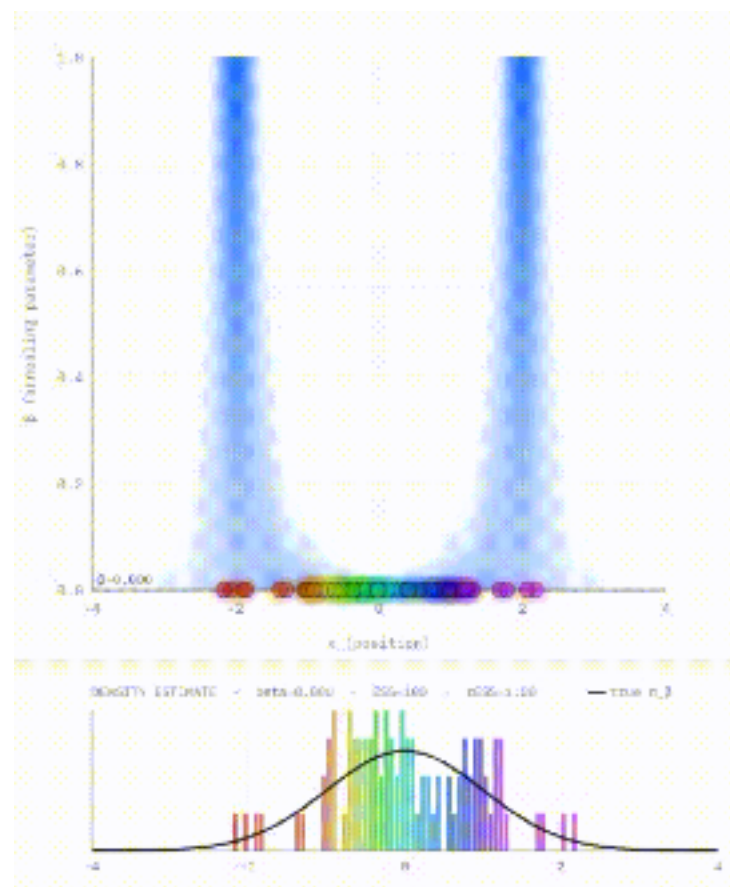


SEQUENTIAL VERSUS PARALLEL ANNEALING

- ▶ We simultaneously evolve a system of particles to approximation the annealing path

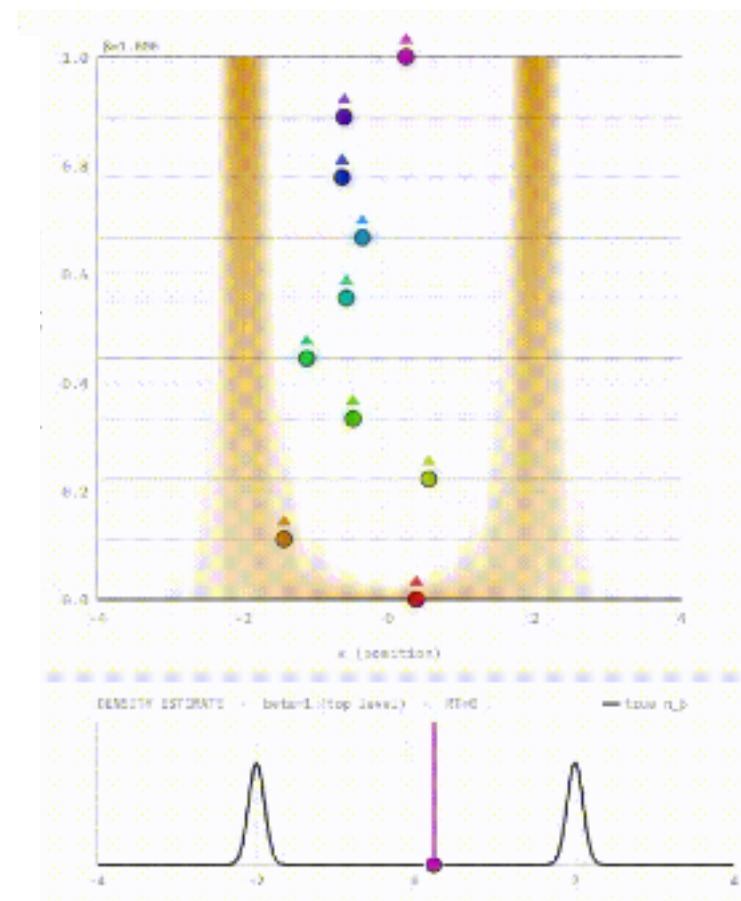
Sequential Annealing

- ▶ Construct path sequentially
- ▶ Samples generated in parallel
- ▶ Built using importance sampling
- ▶ Interacting → Sequential Monte Carlo Samplers
- ▶ Distributed → Simulated sampling



Parallel Annealing

- ▶ Construct path in parallel
- ▶ Samples generated in sequentially
- ▶ Built using MCMC
- ▶ Interacting → Parallel Tempering
- ▶ Distributed → Annealed Importance Sampling



PLAN FOR TERM